Bidding Strategy of a Battery System

46040 Introduction to energy analytics Spring 24

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<u>Group 22</u>

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Task 1: Statistical analysis of historical prices for DK2
Task 2: Battery arbitrage problem
Task 3: Prosumer analytics

Statistical analysis, battery optimization, prosumer benefits analyzed for DK2 area. Profitability trends, efficiency impact, PV system investments discussed.

In this report, all students contributed equally to all parts and are in agreement to its content.

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Introduction

In this report we will dive into an analysis of the energy market for the DK2 area. Through a statistical analysis of historical prices, optimization of battery arbitrage, and evaluation of prosumer analytics, we aim to uncover profitability trends, assess the impact of efficiency on profits, and discuss the potential returns on investments in PV systems.

Task 1: Statistical analysis of historical prices for DK2 Area

1.1

The histogram below displays the average annual electricity spot prices in the DK2 area for 5 years:

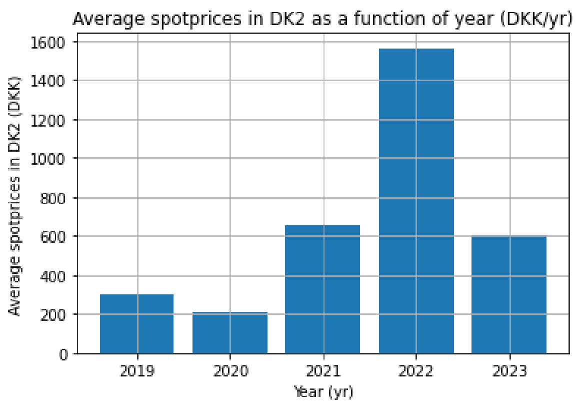

Figure 1: Average spot prices in the DK2 area from 2019 to 2023.

In our analysis of the historical spot price data for the DK2 area from 2019 to 2023, we observe some notable fluctuations in average electricity prices. The data displays a marked decline from 2019 to 2020, followed by a sharp increase in 2021 and a peak in 2022, before declining again in 2023 to approx. 2021-levels. This trend suggests significant volatility in the energy market, possibly due to varying supply and demand dynamics, the shifts in energy policy, or other broader influences like the Russia-Ukraine war, which is the prime suspect. We can underpin this argument of volatility from a statistics point of view. We do it by conducting a one-way ANOVA test, which confirms that the change of mean from year to year is statistically significant, when we have a significance level of 5%. Which means that, it suggests the average spot prices are not equal across the different years.

1.2

The plot in **Figure 2** (below) shows the average spot price per hour of the day for each of the 5 considered years in DK2.

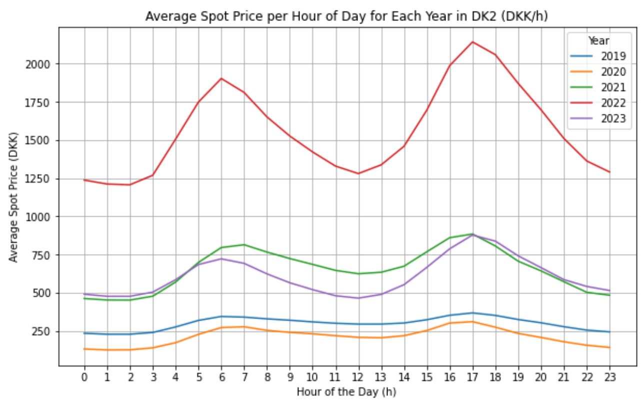

Figure 2: A plot of the spot price pr. hour during the average day for each year considered (2019-2023)

The graph reveals consistent daily patterns, with prices surging during morning and evening hours and then declining at night. This pattern underscores the dynamic relationship between daily demand cycles and the costs of electricity generation.

We see the smallest price surging in 2019 and 2020, where 2019 has the smallest price surges. There are larger price surges in 2023. Followed by even larger, but quite similar surges, in 2021. The largest price surge is clearly in 2022. A similar pattern (though 2021 and 2023 are swapped) is found in the average spot price per year. There is a clear pattern between how large the average spot price is, and the magnitude of the price surges.

In our analysis, 2022 stands out as the year with the most variations between the highest and lowest electricity prices throughout the day. This variability gives us the opportunity to charge the battery when prices are at their lowest and to discharge (or sell the electricity) when prices reach their peak. By doing so, we can leverage the price differences to our advantage, and potentially increase the profits from our battery operations. The greater the price fluctuations, the more incentive there is to engage in battery arbitrage to balance out these differences for personal consumption or for economic gain through buying low and selling high.

To complement our plot in **Figure 2**, we have quantified the price variability for each year using standard deviation calculations.

Year 2019: Standard Deviation = 41.29 DKK
Year 2020: Standard Deviation = 54.88 DKK
Year 2021: Standard Deviation = 132.05 DKK
Year 2022: Standard Deviation = 280.98 DKK
Year 2023: Standard Deviation = 120.07 DKK

These calculations show that the year 2022 had the highest variability, with a standard deviation of 280.98 DKK, indicating the largest daily price change. This aligns with our earlier observations that the significant price swings in 2022 gives the best conditions for battery arbitrage operations. As price fluctuations increase, so does the incentive to use a battery system for arbitrage, which is supported by the price volatility observed in 2022 compared to the lower standard deviations of 41.29 DKK in 2019 and 54.88 DKK in 2020.

Task 2: Battery arbitrage problem

Now we shall look further into the strategic exploitation of a battery energy storage system, solely for the purpose of arbitrage, within the DK2 market area. We consider a system with a 5 kW power capacity and 10 kWh energy capacity, operating with high charging and discharging efficiencies of 99%, and maintaining the battery's state of charge between 10% and 100%.

Our optimization strategy depends on detailed scheduling of the battery's charge and discharge cycles on a daily basis, from midnight to midnight, with the aim of capitalizing on the known spot price fluctuations to buy low and sell high, thereby maximizing profitability.

2.1

On the below figure the aggregated profit per year is plotted as a function of year:

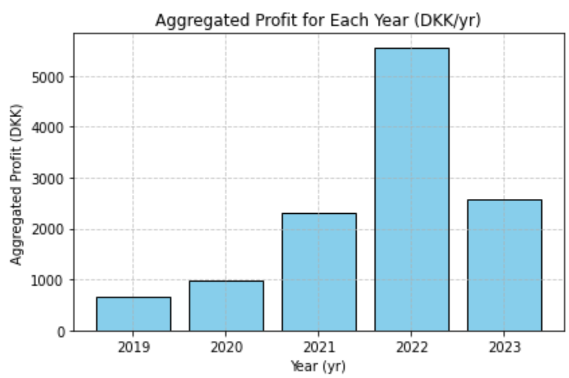

Figure 3: The aggregated profit for the years 2019-2023

Figure 3 indicates that 2022 stands out as the most profitable year. This can be linked to the heightened volatility in spot prices during that year, as identified by our analysis in **1.2**. The broader price differentials provide opportunity for arbitrage, which is optimally exploited through the battery's daily charge and discharge activities.

These conditions (relatively lower lows and higher highs) were less pronounced in other years, making 2022 distinctly favorable for battery arbitrage. Moreover, the optimization function applied in the analysis ensured that the battery's operation was tailored to the unique price patterns of each day, further augmenting the profitability for that year.

2.2

We will now explore the relationship between daily profits and average daily electricity prices.

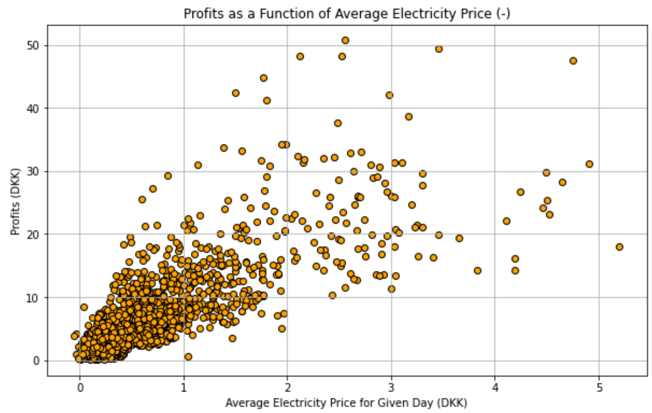

Figure 4 - Profit as a function of the average electricity price.

The scatter plot in **Figure 4** presents a moderate positive correlation, indicating that as the average electricity price increases, so does the profit from battery arbitrage. This correlation coefficient of approximately 0.79 confirms the presence of a significant, yet not perfect, positive linear relationship between the two variables.

However, a more insightful revelation comes from the second scatter plot in **Figure 5**, where we examine the correlation between daily profits and the standard deviation of the daily electricity prices.

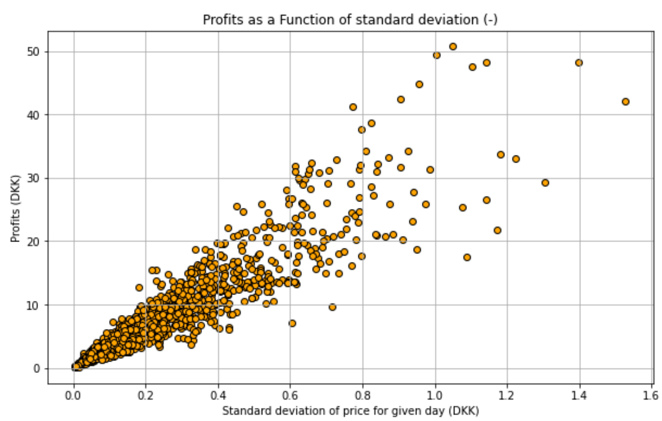

Figure 5 - Daily profits as a function of the standard deviation of the daily electricity prices.

This plot reveals a stronger positive correlation, with a coefficient of around 0.93, suggesting that the variability in daily prices, rather than the average price itself, has a more substantial impact on the potential for arbitrage profits.

This stronger correlation with price fluctuation implies that greater price variability within a day enhances the opportunities for battery arbitrage, as the battery can be charged when prices dip and discharged when they spike, leading to higher profits, as there are either more opportunities to make money, larger gaps to take advantage of or both.

In conclusion, while average prices provide some indication of arbitrage opportunities, it is the extent of price fluctuation throughout the day that is more determinative of the profitability of battery arbitrage operations.

2.3

Let's take a look into the impact of charging and discharging efficiencies (ηc and ηd) on the profitability of battery arbitrage across the five years, with efficiencies set at 99%, 95%, and 90%.

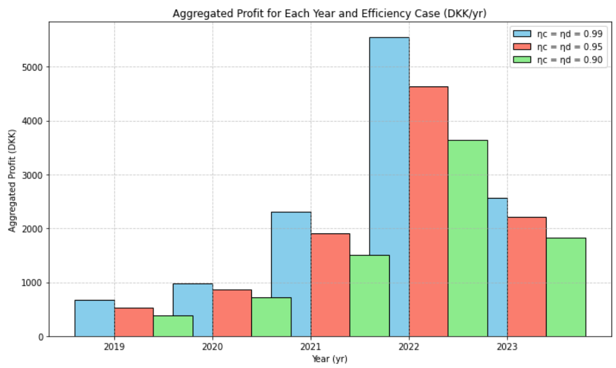

Figure 6 - The aggregated profit for each year and for each of the three efficiency cases.

The bar chart in **Figure 6** illustrates a clear trend: higher efficiencies correlate with increased profits. This is particularly noticeable in the year 2022, because it has the highest aggregated profit.

We have made a linear regression of the relationship between the efficiency, and the profit for every year. This analysis yields a slope of 89.16. Which means that when the efficiency changes 1 percentage point, the profit changes by 89 DKK/yr.

We have done the same for the relationship between the average spot price for a year, and the profits seen. Where we do this for all three efficiency. This analysis yields a slope of 0.83. This means that when the average spot price for a year changes by 1 DKK then the profit changes by 0.83 DKK/yr.

Looking back at this analysis, one would probably not try to do a linear regression on the efficiency. One could imagine it would have a nonlinear relationship.

But, we still conclude with confidence that the efficiency has a much larger effect on the profits than the spot price. Though, high variability in the spot price (which we have seen hand in hand with high spot prices in this report) AND a high efficiency are both instrumental in making profits.

Task 3: Prosumer analytics

We now consider a prosumer, again located in DK2. Hourly PV generation and consumption values are given by the ProsumerHourly csv file. Consider only years 2022 and 2023. The prosumer buys energy at the Buy price and sells at the Sell price at each hour; you have calculated these prices in a previous hands-on (refer to the previous hands-on exercises). The prosumer is subject to hourly net metering, i.e., at each hour the prosumer pays for the energy import or receives money for the energy export.

3.1

Setting up a program using the supplied in UsefulFunctions.py file, we were able to investigate the consumption and costs of the prosumer in 2022 and 2023. Total costs:

- 2022: 10225,29 DKK
- 2023: 8241,11 DKK

A rough calculation based on average values was then calculated, resulting in respectively 9963 DKK and 8038 DKK, which was close, but not perfect. The reason for this difference was that the real cost calculation are based on the actual hourly consumption and corresponding buy prices, while the rough calculation method estimates costs by multiplying the total consumption for the year with the average buy price for that year. If we should try and dive deeper into a reason why an average would deviate with this much (approx 2-3% from the real cost):

- Hourly variation: The first method considers actual hourly consumption and corresponding buy prices, capturing fluctuations accurately.
- Tariffs and pricing structures: It accounts for variations in tariffs or pricing structures, reflecting actual costs incurred.
- Influence of outliers: Hourly variations, especially outliers, significantly impact total costs in the first method.
- Accuracy of averages: The rough method's accuracy depends on the representativeness of the average buy price used, potentially leading to discrepancies.

As to the last two points highlight, it is difficult to model something precisely due to outliers and other discrepancies, which is an important factor to consider when it concerns humans and their consumption. Like in the lecture where an investigation of trying to model housing prices showed the same difficulty, due to unpredictability of humans, e.g. buying or selling at irrational prices.

3.2

The total profit with PV (netting of one hour) for 2022 and 2023 account respectively for 501.85589 DKK and -3294.59 DKK, while the total costs without PV account for 10225.29 DKK in 2022 and 8241.11 DKK in 2023. This information is presented in the left hand of Table 1, where the difference between the money spent with and without PV for each respective year (2022 and 2023) is calculated, with an average difference of 7836.83295 DKK. On the right side we present a simple calculation for the expected return of investment (ROI) with PV in a time period of 20 years, with an assumed capital expenditure (CAPEX) of 80,000 DKK and an assumed operational expenditure (OPEX) of 500 DKK/year. In this manner, we obtain a ROI of 74%.

		Money *spend* on electricity				20 Years			
Net	**Year**	**With PV**	**No PV** (i.e. just consump.)	**Difference** (i.e. money saved)	**Netting**	**Money saved**	**W. CAPEX (80k)**	**+W OPEX (500 DKK/year)**	**ROI (Percentage)**
Hourly	2022	-501,85589	10225,29	10727,1459	Hour	156736,7	76736,6589	66736,6589	74%
	2023	3294,59	8241,11	4946,52					
	Average:			7836,83295					

Table 1: Money spent on electricity with and without PV.
Please note assumptions used, especially for CAPEX and OPEX

A more precise method to determine if the PV is a good 20-year investment can be done through the calculation of the net present value (NPV).

$$NPV = -C_0 + \sum_{t=1}^{20} \frac{C \cdot F_t}{(1+r)^t} \qquad (1)$$

Hence, assuming a cost of the PV system in DKK of 79478.64 DKK, a maintenance cost and energy generation in kWh per year of 500 and 7000 DKK respectively, initial electricity price of 2 DKK/kWh expected to increase a 2% annually, and a 5% discount rate; we obtain a positive generated Net Present Value (NPV) of the PV system over 20 years of 123390.58 DKK, suggesting the value added of the investment considering the presented initial investment, energy savings, maintenance costs, and discount rate.

3.3

Now assume that the prosumer buys the battery described in Task 2. Your charging and discharging efficiencies are equal to 95%. Optimize the operation of the battery for every day of 2022 and 2023, with the goal of minimizing prosumer costs under net metering. What is the benefit that the battery brings to the prosumer on a yearly basis? The comparison should be made between the case where the prosumer owns only the PV and the case where both the PV and a battery are present.

The benefit brought to the prosumer on a yearly basis, comparing the case where the prosumer owns only the PV and the case where both the PV and a battery are present:

- 2022: 5377,55 DKK
- 2023: -3108,84 DKK

I.e. it made more sense, in total, to have the battery available, but was not in 2023 an advantage. Although looking at this critically it does not make sense that the battery doesnt have a positive impact, seing as we use only one hour net metering. Weassume there is a mistake in the code, but this could not be discovered unfortunately.

3.4

Considering the one hour net metering, it would definitely be a recommendation to have a battery installed. More realistically in Denmark's current situation with no net metering, even more. Online the cost of a tatal PV/Battery system varied from 60-120.000 DKK, due to various qualities of each component and situation, but in these years a lot of development in the storage technologies is underway, and considering adding the storage solution to your PV, is a consideration an owner should take.